Executive Summary

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01 FINANCIAL HEALTH SCORE

02 ENGAGEMENT SCORE

03 CUSTOMER SEGMENTATION

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Business Understanding & Overview

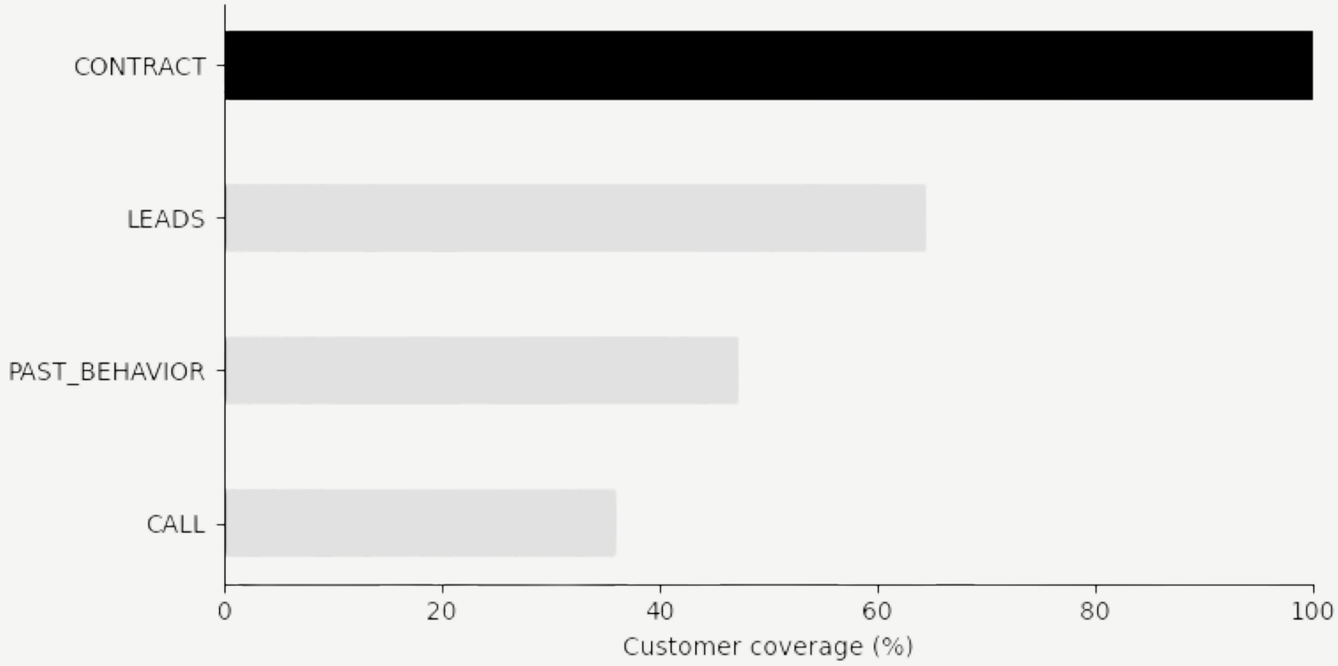
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The company lacks a unified view of customer financial health, engagement, and future potential, limiting its ability to identify risks and deliver personalized actions.

1,000,528
customers

17,791,122
raw records

2009 - 2025
contract history



CONTRACT is the only transactional table covering 100% of customers

Note. While financial records span a long history and cover all customers, engagement data remains fragmented, highlighting the need for a unified customer view.

Data Preparation

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2M duplicated rows	identified in CONTRACT and LEADS.
74.4% of calls	occurred after the June 4, 2025 snapshot.

Key actions

Deduplication
Remove duplicate contracts and leads, retaining the most complete records.

Repair & Engineer Features
Remove low-quality fields, and derive age and contract indicators.

Standardize fields
Unify product codes into common groups, and correct monetary units.

Flag Temporal Risks
Flag post-snapshot calls to enable time-consistent analysis.

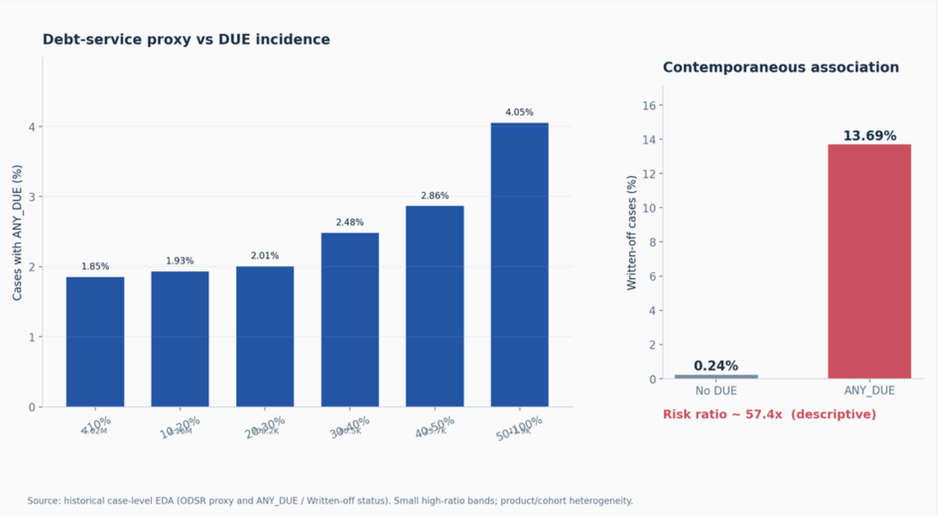
Key Factors Informing the Financial Health Score

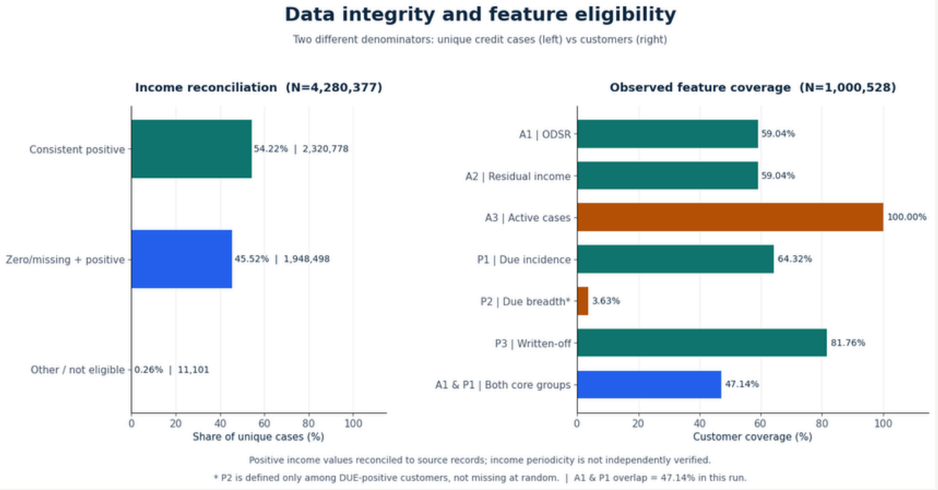
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Debt-service pressure – A1
Case-level DUE: 1.85% to 4.05% across bands; client-level reversal.

Residual cash-flow proxy – A2
Incremental AP vs A1: +0.00098 (small gain).

Payment incident frequency – P1
Any DUE / WO association RR=57.4; same extract, not temporal.





Customer Financial Health Scoring Method

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Base-score | Non-written-off customers

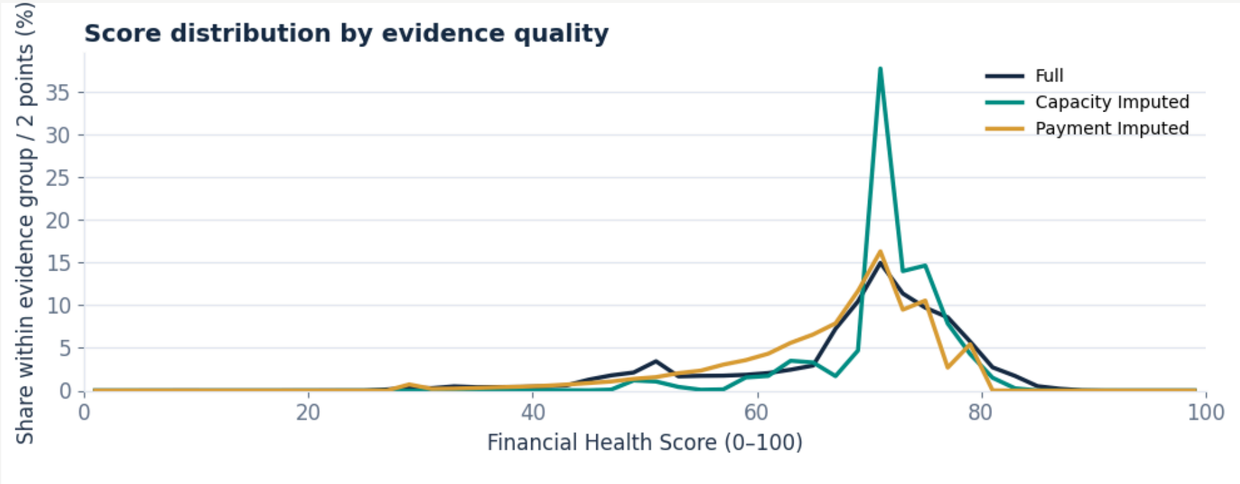
$$FHS_{base} = 0.4\,A1 + 0.1\,A2 + 0.5\,P1$$

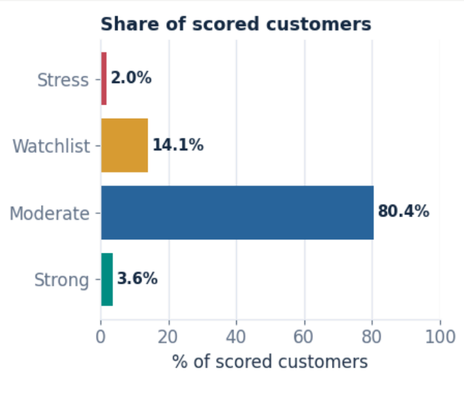
Written-off customers

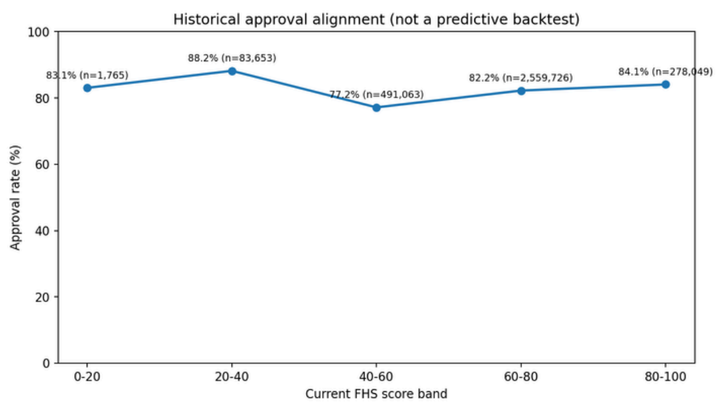
$$FHS = \min(50, FHS_{base})$$

Score components

A1: Observed Debt Service Ratio
A2: Residual Income Proxy
P1: Delinquency Incidence Rate

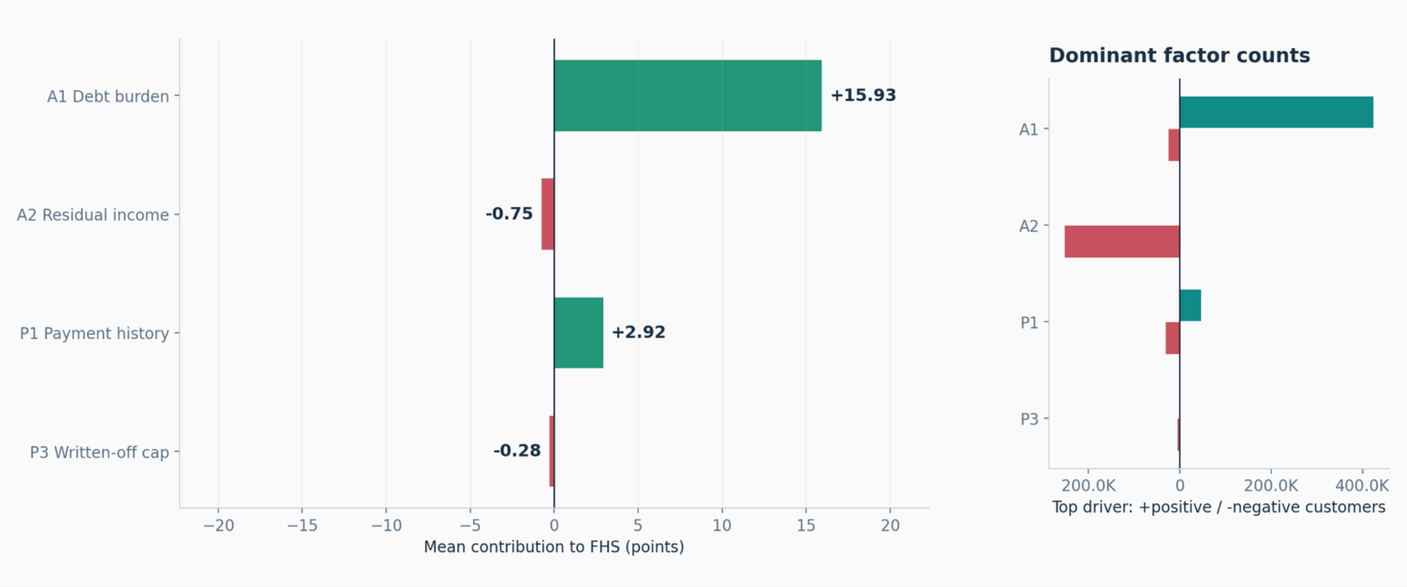






Key factor on Financial Health

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ECONOMIC INTERPRETATION

A1
Lower internal debt burden
+ / − 4 pts per 10 A1 pts

A2
Higher residual income
+ / − 1 pt per 10 A2 pts

P1
Fewer observed DUE flags
+ / − 5 pts per 10 P1 pts

P3
Severe credit event
Cap ≤ 50, not a fixed weight

13,319 WO customers flagged

Key Takeaway.

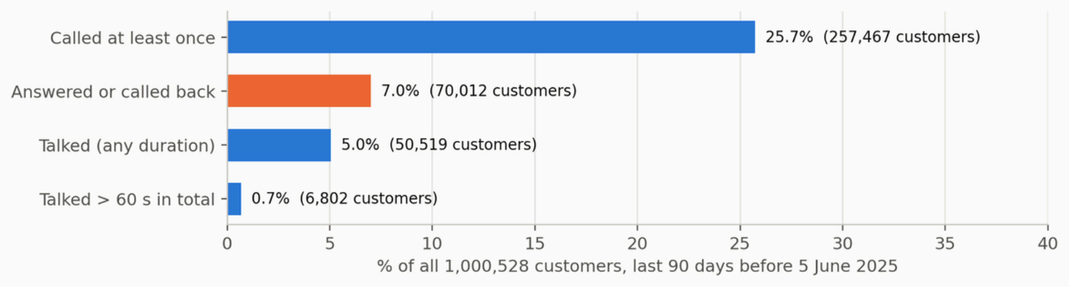
A customer's financial health depends on both repayment capacity and payment behavior. Strong affordability alone does not guarantee strong financial health.

Behind a Trustworthy Engagement Score

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A reliable Engagement Score requires strict time alignment, behavior consolidation, and leakage control across multiple interaction datasets

Why Engagement Score is Needed



Temporal Leakage: 74.4% of Calls Occurred After the Snapshot and were excluded from historical scoring → exclude

Interaction Gap: 25.7% called → 7.0% responded → 0.7% talked over 60s

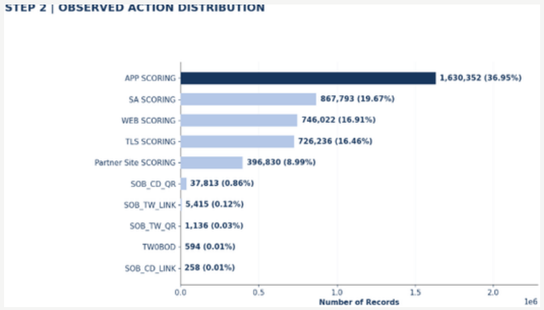
Others: 85.3% missing dispositions, 86.4% zero-duration calls & 4.3% OWN-coded calls

From Noisy Records to Reliable Engagement Signals

Snapshot → Business Filter → Behavioral Signals → Customer-Level Data

Outreach ≠ Engagement. Call frequency measures company effort, not customer interest

Timestamp Quality Metric	Count	Share (%
Raw Events	4,412,617	100.00
Valid Timestamps	4,412,617	100.00
Missing / Unparseable Timestamps	0	0.00



Sessionization Metric	Count
Raw Events	4,412,617
Total Sessions	4,240,013
Multi-event Sessions	141,235
Events Preserved	4,412,617

Session Key = SKP_CLIENT + ACTION + PRODUCT_CODE.
New Session = first valid event, missing/uncomparable gap, or rolling inactivity gap > 300 seconds.
Consecutive gap <= 300 seconds preserves the current session. Missing time/key events form singleton sessions.
Conservation: Sum(Session Event_Count) = Raw Events. No event is deleted.

P0 Audit Category	Actual Value
1. Total Raw Rows in CALL.csv	4,000,000
2. Duplicated CALL_ID Records Removed	12,527
3. Successfully Linked Customer Records	3,987,473
4. Orphaned Records (Not in Customer base)	0

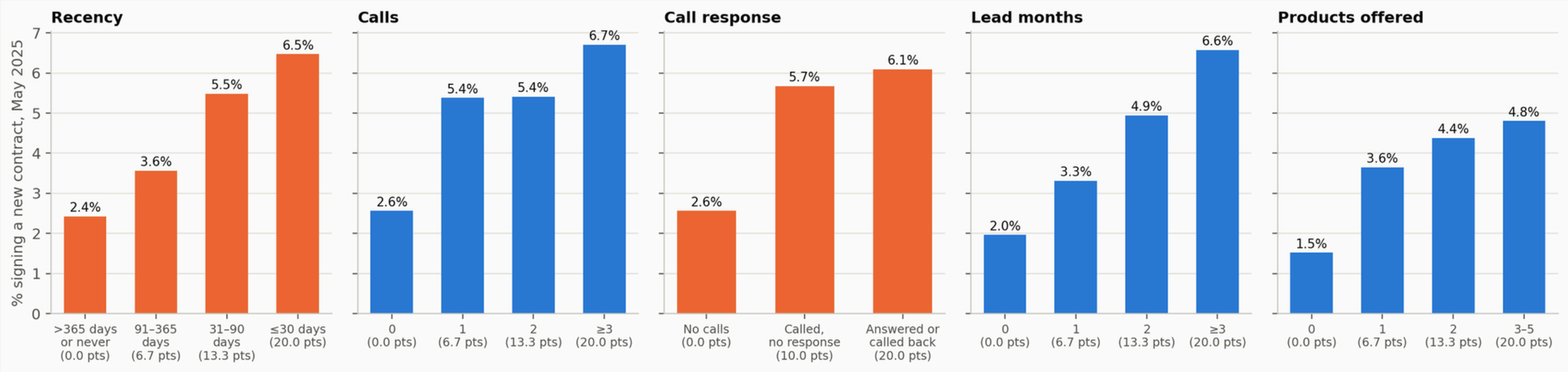
Classification of DURATIONS = 0 Records	Quantity
1. Total Records with DURATIONS = 0	3,451,676
2. False Positive Errors (Connected label but DURATIONS = 0s)	0
3. Missing Staff Processing Errors (OFF_BY is NULL/Empty)	0
4. Total System Error Records Excluded	0
5. Valid 0-second Records (System Recorded Not Connected / Failed)	3,451,676

Five Behavioral Signals, One Explainable Engagement Score

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A transparent 0–100 score achieved 0.670 AUC, just 0.008 below logistic regression (0.678) on the development sample.

Feature Selection: 16 Candidates → 5 Behavioral Signals



Transparent Scoring: *Five behavioral indicators contribute up to 20 points each, producing an auditable 0–100 score that distinguishes company outreach from customer response.*

ENGAGEMENT SCORE = REACH + RESPONSE

$$S_{\text{Eng}} = \underbrace{S_{\text{Calls}} + S_{\text{Leads}} + S_{\text{Products}}}_{0\text{–}60} + \underbrace{S_{\text{Recency}} + S_{\text{CallResponse}}}_{0\text{–}40}$$

ach feature: 0–20 pts · 4-level features: 0, 6.7, 13.3, 20 · Call response: 0, 10, 20

Feature	0 pts	Intermediate tiers	20 pts
Recency	>365 days / never	91–365 / 31–90 days	1–30 days
Calls	0	1 / 2	≥3
Call Response	No calls	Called, no response (10 pts)	Answered / inbound
Lead Months	0	1 / 2	≥3
Products Offered	0	1 / 2	3–5

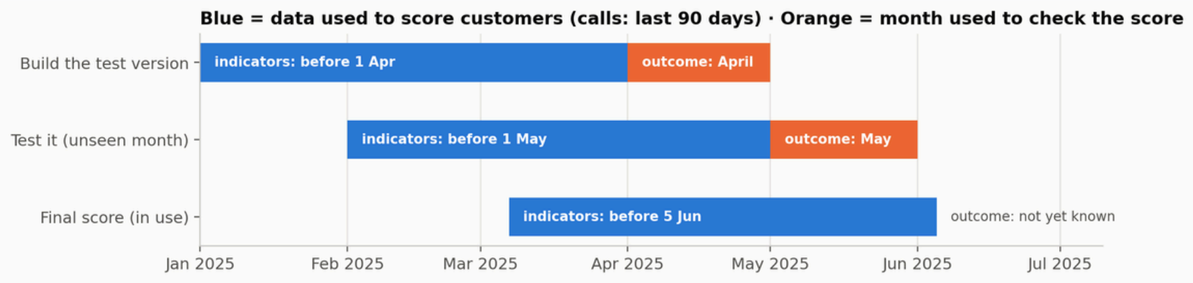
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Does the Score Actually Work?

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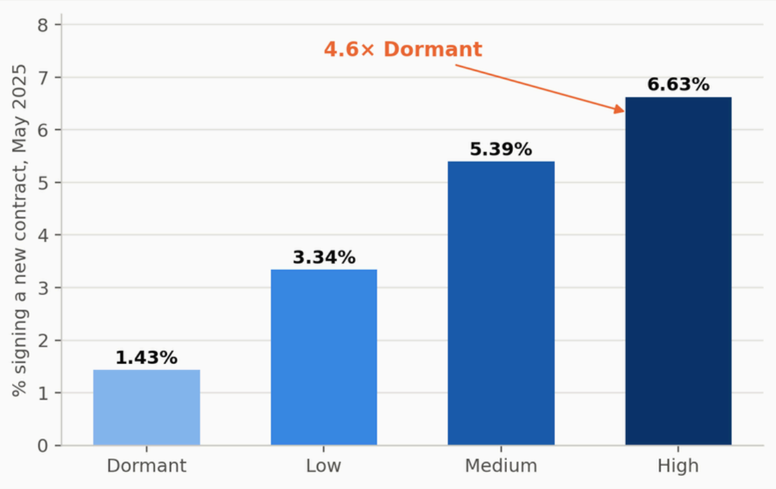
Out-of-time validation achieved 0.670 AUC, with the High Engagement group generating 30.1% of new contract value from just 12.1% of customers.

Leakage-Free Temporal Validation



Out-of-Time Validation: *The scoring methodology was rebuilt using April outcomes and pre-April indicators, then evaluated against unseen May 2025 contracts using only pre-May features, preventing future-outcome leakage.*

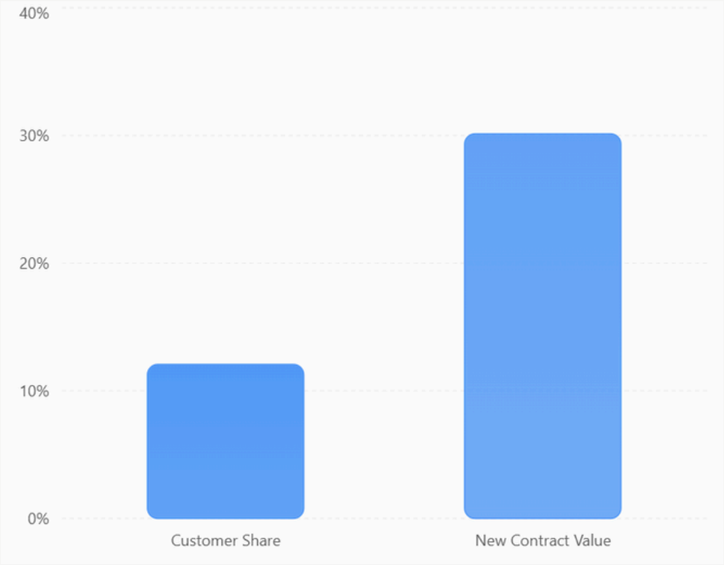
Engagement Score Predictive Separation



Monotonic Risk Ranking: May contract-signing rates rose consistently from Dormant (1.43%) to Low (3.34%), Medium (5.39%) and High (6.63%), representing a 4.6× lift between the extreme groups.

Business Value Concentration: High-engagement customers represented only 12.1% of the population but accounted for 30.1% of newly signed contract value — approximately 2.5× their customer share.

High Engagement: Customer Share vs. Contract Value

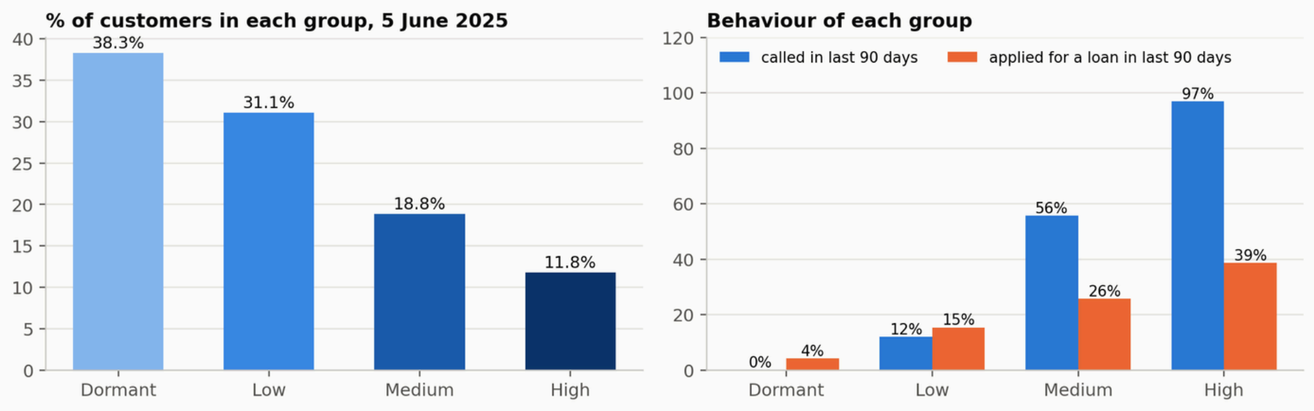


Engagement is not only a ranking metric

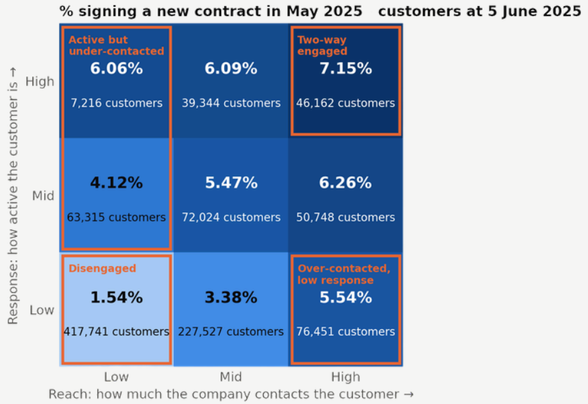
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Two-way engaged customers recorded a 7.15% contract-signing rate versus just 1.54% among disengaged customers — a 4.6× difference.

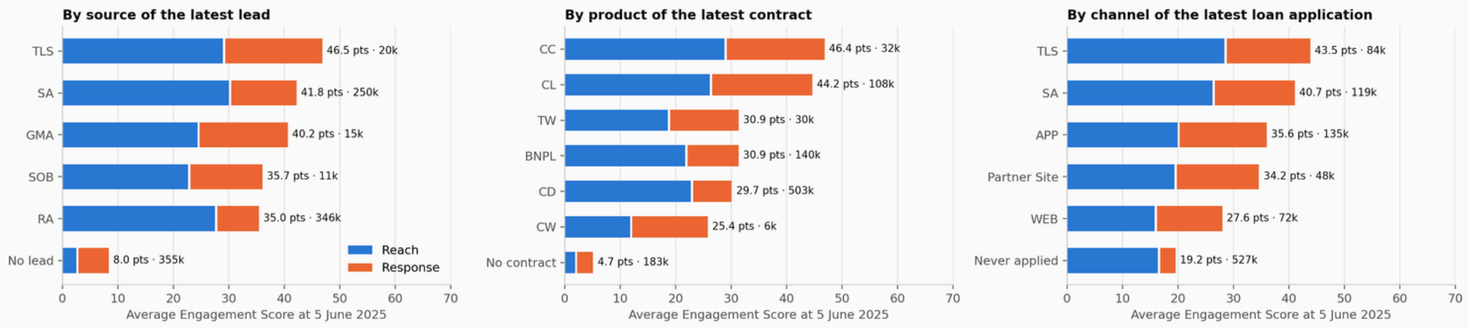
Who Are Our Customers?



Customer Segmentation: 69.4% of customers fall into Dormant or Low Engagement, while High-engagement customers show substantially greater loan-application activity (39% vs. 4% among Dormant).



What Distinguishes Engagement Levels?



Engagement Differentiators: Customers with TLS-sourced leads average 46.5 engagement points versus 8.0 for those without leads; CC contract holders average 46.4 versus 4.7 for customers without contracts.

Actionable Opportunities: Two-way engaged customers show the highest signing rate (7.15%; 46,162 customers), while 76,451 over-contacted, low-response customers warrant contact-strategy optimization.

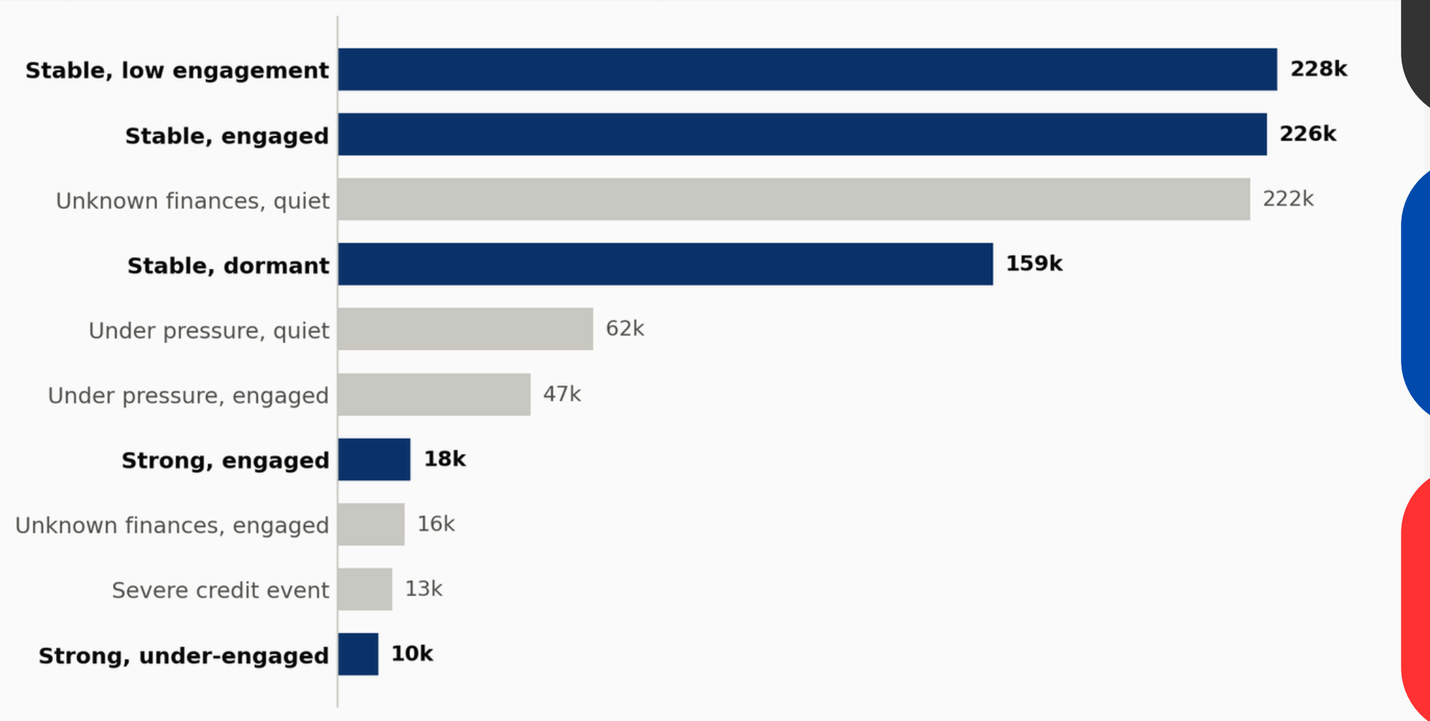
Recommended Actions: Prioritize two-way engaged customers for conversion, expand targeted outreach to active but under-contacted customers, and reassess excessive contact attempts.

High engagement drives value, but financial health determines the action

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Customer Distribution Across 10 Segments

Figure 1. Customer coverage by table



38.7% **Stable but dormant**
→ Prioritize reactivation campaigns

22.6% **Stable & engaged**
→ Target personalized product offers

4.7% **Engaged but under financial pressure**
→ Review risk

Segmentation Method. FHS cutoffs: 40/60/80 | Engagement Score cutoffs: 20/40/60

Recommendation Actions

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Segment	Next Action	Why This Action?
Severe Credit Event	Credit review; no new offers	Written-off history; 87.8% rejected in 180 days.
Unknown, Engaged	Collect financial information	FHS unavailable, but 36.0% applied in the last 90 days.
Unknown, Quiet	Low-cost data update outreach	64.5% recently rejected; only 0.63% signed in May.
Under Pressure, Engaged	Debt counseling; pause cross-sell	FHS < 60; 96.9% hold active loans; 19.5% of May's new contract value.
Under Pressure, Quiet	Monitor and send repayment reminders	Low engagement (18.5), while 78.6% still have active loans.
Stable, Engaged	Personalized product offers	88.5% have unsigned product interests; 38.4% of May's new contract value.
Stable, Low Engagement	Digital nurturing campaigns	84.8% have unsigned interests, but only 12.8% recently applied.
Stable, Dormant	Low-cost reactivation	Only 4.6% recently applied; May signing rate: 2.29%.
Strong, Engaged	Prioritize offers / limit review	Strong financial health; 90.6% have unsigned interests.
Strong, Under-engaged	Proactive calls	Strong finances but low outreach score (17.3/60).